# Status of language on frozen LeWM for MMAGAI

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**Abstract**

LeWorldModel stays frozen. A linear readout already recovers Push-T block angle, including after a half turn. The planner that scores 90 percent on a dataset goal image does not spin the block in place, even when it is shown a rendered frame of that spin. Language for this course should name goals that planner already reaches.

## 1 Introduction

The model is the public Push-T checkpoint of LeWorldModel [Maes et al., 2026]. No language model sits inside the JEPA, and text is not turned into pixels. This note records which interfaces were measured on a CPU Xeon and on one rented RTX 4090, and which course step follows.

## 2 Methods

Angle was probed on the Xeon by a ridge map from the 192-dimensional latent to $(\cos\theta, \sin\theta)$, with episodes held out. A separate least-squares map sent each of four command strings to one latent displacement. On the 4090, cross-entropy planning used 300 samples, 30 iterations, 30 elites, horizon 5, and an action block of 5. Actions were z-scored with the Push-T training column before the action encoder, then inverse-transformed and clipped to $[-1, 1]$ before the environment step. The image-goal task took its start and its goal from dataset rows 25 steps apart, with a budget of 50 environment steps. A second task rendered the same block position at $\theta + 90°$ or $\theta + 180°$ and used that frame as the goal latent. The rented machine is no longer available. Nothing below was recomputed for this note.

## 3 Results

The CPU ridge has held-out circular mean absolute error 12.08 degrees on scripted 180° frames and 14.58 degrees on frames within 30 degrees of angle 0. The decision from that comparison is to leave the encoder frozen. A

Table 1: Measured interfaces. Reach is a fraction of episodes.

| Measurement | Value |
|---|---|
| Ridge MAE, 180° frames | 12.08 degrees |
| Ridge MAE, within 30 degrees of 0 | 14.58 degrees |
| Constant $\Delta z$ residual ratio | 0.991 |
| Image-goal CEM | 45/50 (90 percent) |
| Guessed $\times 5$ CEM | 0/50 (invalid) |
| Probe-cost reach, +90° / 180° | 0.10 / 0.00 |
| Oracle rendered goal, +90°, budget 50 | 0.00 |
| Oracle rendered goal, +180°, budgets 50 and 100 | 0.00 |

constant displacement per command has residual ratio 0.991 against the root-mean-square of the observed latent change. Official image-goal CEM succeeds on 45 of 50 episodes, which is 90 percent. The action column used for that run has mean $[-0.00781, 0.00686]$ and unbiased standard deviation $[0.20847, 0.20675]$. An earlier planner that multiplied actions by 5, without those statistics, succeeded on 0 of 50 episodes and is not a valid baseline. A cost written on the angle probe, with the official action transform, reaches a +90° turn on fraction 0.10 of 20 held-out starts and a 180° turn on fraction 0.00. Giving CEM the encoding of a rendered in-place target does not raise that rate. Reach is 0.00 for +90° at budget 50, and 0.00 for +180° at budgets 50 and 100. Table 1 collects these figures.

## 4 Discussion

The planner can chase a dataset goal image. It cannot spin the T in place. A command list for the course should name goals the planner already reaches, such as putting the block on the target or other expert-style goals from the demonstration set. It should not ask for a 180° in-place skill. Captioning or indexing those reachable goals from a small command list is the next implementation step. Hierarchical planning is future work, not this checkpoint's missing text encoder.

## 5 Conclusion

The encoder stays frozen because angle is already linearly readable. Language is useful here as an index into goal images the existing CEM can follow. In-place half-turns are outside that set under the horizon and budget that were tested.

## References

Lucas Maes, Quentin Le Lidec, Damien Scieur, Yann LeCun, and Randall Balestriero. Leworldmodel: Stable end-to-end joint-embedding predictive architecture from pixels. *arXiv preprint arXiv:2603.19312*, 2026. URL https://arxiv.org/abs/2603.19312.